

Name: _____ ID: _____

Mid-sem exam – CS F407-Artificial Intelligence
Weightage: 25%
Maximum Marks: 75

Exam type: Closed Book
Date : 8th Oct 2025
Time: 90 minutes

Question	1	2	3	4	5	6	7	8	9	Total
Marks							x	x	x	
Remarks							x	x	x	

General instruction:

- Please read the question carefully before attempting it and be precise with the answers and use the space for answer wisely. **No extra space will be provided.**
- Numerical type questions no partial marks only if the total step is correct, marks will be awarded.

Q1. Suppose a genetic algorithm uses chromosomes of the form $x = abcdefgh$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as: $f(x) = (a+b)-(c+d)+(e+f)-(g+h)$, and let the initial population consist of four individuals with the following chromosomes: $x_1 = 65413532$ $x_2 = 87126601$ $x_3 = 23921285$ $x_4 = 41852094$ **(20 marks)**

a) Evaluate the fitness of each individual, showing all your workings, and arrange them in order with the fittest first and the least fit last. **[4 marks]**

Answer: $f(x_1) = (6+5)-(4+1)+(3+5)-(3+2)=9$ $f(x_2) = (8+7)-(1+2)+(6+6)-(0+1)=23$ $f(x_3) = (2+3)-(9+2)+(1+2)-(8+5)=-16$ $f(x_4) = (4+1)-(8+5)+(2+0)-(9+4)=-19$ The order is x_2 , x_1 , x_3 and x_4 .

b) Perform the following crossover operations: **[2x3=6 marks]**

i) Cross the fittest two individuals using one-point crossover at the middle point.

ii) Cross the second and third fittest individuals using a two-point crossover (points b and f).

iii) Cross the first and third fittest individuals (ranked 1st and 3rd) using a uniform crossover.

Answer:

i) One-point crossover on x_2 and x_1 :

$x_2 = 8712\ 6601$ $x_1 = 6541\ 3532 \Rightarrow O_1 = 87123532$ $O_2 = 65416601$

ii) Two-point crossover on x_1 and x_3 $x_1 = 65\ 4135\ 32$ $x_3 = 23\ 9212\ 85 \Rightarrow O_3 = 65921232$ $O_4 = 23413585$

iii) In the simplest case uniform crossover means just a random exchange of genes between two parents. For example, we may swap genes at positions a, d and f of parents x_2 and x_3 : $x_2 = 87126601$ $x_3 = 23921285 \Rightarrow O_5 = 27126201$ $O_6 = 83921685$

(c) Suppose the new population consists of the six offspring individuals received by the crossover operations in the above question. Evaluate the fitness of the new population, showing all your workings. Has the overall fitness improved? **[6 marks]**

Answer: The new population is: $O_1 = 87123532$ $O_2 = 65416601$ $O_3 = 65921232$ $O_4 = 23413585$ $O_5 = 27126201$ $O_6 = 83921685$ Now apply the fitness function $f(x) = (a+b)-(c+d)+(e+f)-(g+h)$: $f(O_1) = (8+7)-(1+2)+(3+5)-(3+2)=15$ $f(O_2) = (6+5)-(4+1)+(6+6)-(0+1)=17$ $f(O_3) = (6+5)-(9+2)+(1+2)-(3+2)=-2$ $f(O_4) =$

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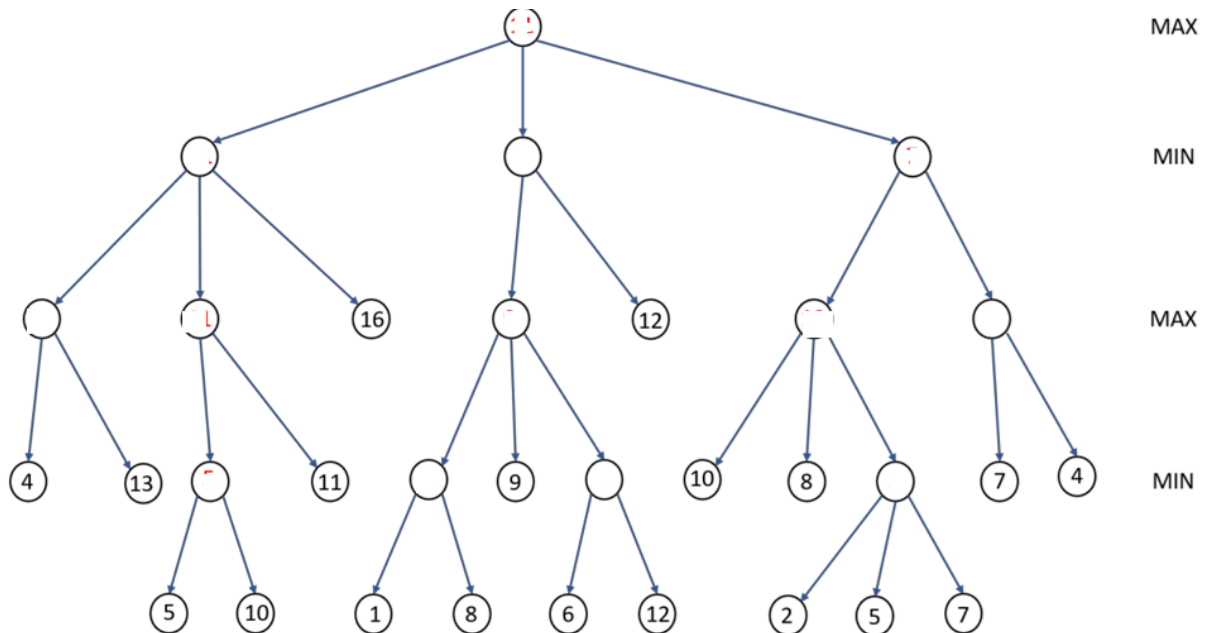
$= (2+3)-(4+1)+(3+5)-(8+5)=-5$ $f(O5) = (2+7)-(1+2)+(6+2)-(0+1)=13$ $f(O6) = (8+3)-(9+2)+(1+6)-(8+5)=-6$ The overall fitness has improved

(d) By looking at the fitness function and considering that genes can only be digits between 0 and 9 find the chromosome representing the optimal solution (i.e. with the maximum fitness). Find the value of the maximum fitness. [4 marks]

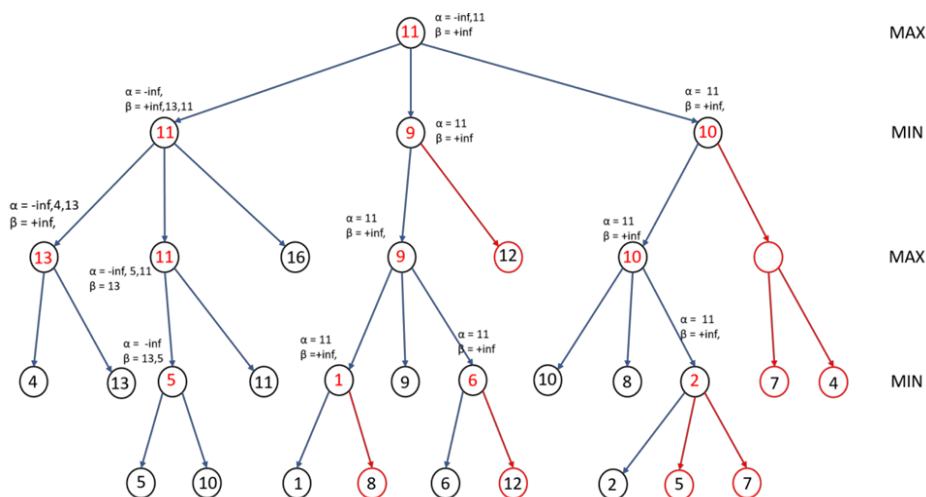
Answer: The optimal solution should have a chromosome that gives the maximum of the fitness function $\max f(x) = \max[(a+b)-(c+d)+(e+f)-(g+h)]$. Because genes can only be digits from 0 to 9, the optimal solution should be: $x_{\text{optimal}} = 99009900$, and the maximum fitness is $f(x_{\text{optimal}}) = (9+9)-(0+0)+(9+9)-(0+0) = 36$

Q2: (a) Use the Minimax algorithm to compute the minimax value at each node for the game tree below.

(b) Use the Alpha-Beta pruning algorithm to prune the game tree in Problem 1 (a) assuming child nodes are visited from left to right. Show all final alpha and beta values computed at root, each internal node explored, and at the top of pruned branches. [10+10= 20 marks]



Answer:



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Q3. Tick mark on the correct option. (2.5x4= 10 marks) **c d b d**

(a) What is the correct translation of the following statement into mathematical logic? "Some real numbers are rational"

- (A) $\exists x (\text{real}(x) \vee \text{rational}(x))$
- (B) $\forall x (\text{real}(x) \rightarrow \text{rational}(x))$
- (C) $\exists x (\text{real}(x) \wedge \text{rational}(x))$
- (D) $\exists x (\text{rational}(x) \rightarrow \text{real}(x))$

(b) Which one of the following options is CORRECT given three positive integers x, y and z, and a predicate?

$$P(x) = \neg(x=1) \wedge \forall y (\exists z (x=y*z) \Rightarrow (y=x) \vee (y=1))$$

- A. P(x) being true means that x is a number other than 1
- B. P(x) is always true irrespective of the value of x
- C. P(x) being true means that x has exactly two factors other than 1 and x
- D. P(x) being true means that x is a prime number

(c) Suppose the predicate F(x, y, t) is used to represent the statement that person x can fool person y at time t. which one of the statements below expresses best the meaning of the formula $\forall x \exists y \exists t (\neg F(x, y, t))$?

- a. Everyone can fool some person at some time
- b. No one can fool everyone all the time
- c. Everyone cannot fool some person all the time
- d. No one can fool some person at some time

(d) Which one of the following is the most appropriate logical formula to represent the statement? "Gold and silver ornaments are precious". The following notations are used: G(x): x is a gold ornament S(x): x is a silver ornament P(x): x is precious

- a. $\forall x (P(x) \rightarrow (G(x) \wedge S(x)))$
- b. $\forall x ((G(x) \wedge S(x)) \rightarrow P(x))$
- c. $\exists x ((G(x) \wedge S(x)) \rightarrow P(x))$
- d. $\forall x ((G(x) \vee S(x)) \rightarrow P(x))$

Q4. Tick mark the correct order for the algorithm implemented on the given graph consider 0 as root node. [2.5x4=10 marks]. **All a**

(a) DFS

I. 0135264 II.. 0123456 III. 0132456 IV.. 0123465

(b) BFS

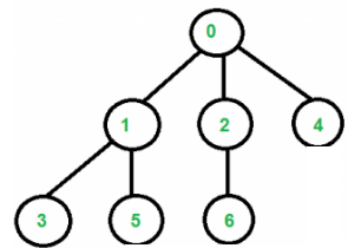
I, . 0124356 II..0135264 III. 0132456 IV.. 0123465

(c) Iterative Deepening Depth First Search at level 2

I. 0135264 II.. 0123456 III. 0132456 IV.. 0123465

(d) Depth limited search D=1

I. 0124356 II..0135264 III. 0132456 IV.. 0123465



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Q5: Imagine that a student wants to hand out fliers about an upcoming programming contest. The student wants to visit the Memorial Union (M), Wisconsin Institute of Discovery (W), Computer Sciences Building (S), and Engineering Hall (E) to deliver the fliers. The goal is to find a tour as short as possible. The distance matrix between these locations is given as follows: **[2+ 3+ 4+ 3=15marks]**:

	M	W	E	S
M	0	1.1	1.4	0.9
W	1.1	0	0.6	0.7
E	1.4	0.6	0	0.5
S	0.9	0.7	0.5	0

[a] The student starts applying a hill-climbing algorithm from the initial state: <W-M-E-S>. Compute total distance of current state.

[Ans] $d(W,M)+d(M,E)+d(E,S) = 1.1 + 1.4 + 0.5 = 3.0$

[b] What are the possible neighboring states of <W-M-E-S> and the total distances?

[Ans] : <M-W-E-S> : $1.1 + 0.6 + 0.5 = 2.2$ <E-M-W-S> : $1.4 + 1.1 + 0.7 = 3.2$ <S-M-E-W> : $0.9 + 1.4 + 0.6 = 2.9$ <W-E-M-S> : $0.6 + 1.4 + 0.9 = 2.9$ <W-S-E-M> : $0.7 + 0.5 + 1.4 = 2.6$ <W-M-S-E> : $1.1 + 0.9 + 0.5 = 2.5$

[c] What is the next state reached by hill-climbing? Or explain why there is no next state, if there isn't one?
[Ans] : <M-W-E-S>

[d] If c) had a next state iterate further until the algorithm terminates and list out the set of states from initial state to final state.

[Ans] <W-M-E-S> $\rightarrow$ <M-W-E-S> $\rightarrow$ <M-S-E-W>

[e] What is the total distance of the final state?

[Ans] distance = 2.0

Rough work

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